

Activation Matters: Test-time Activated Negative Labels for OOD Detection with Vision-Language Models

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Abstract

Out-of-distribution (OOD) detection aims to identify samples that deviate from in-distribution (ID). One popular pipeline addresses this by introducing negative labels distant from ID classes and detecting OOD based on their distance to these labels. However, such labels may present poor activation on OOD samples, failing to capture the OOD characteristics. To address this, we propose Test-time Activated Negative Labels (TANL) by dynamically evaluating activation levels across the corpus dataset and mining candidate labels with high activation responses during the testing process. Specifically, TANL identifies high-confidence test images online and accumulates their assignment probabilities over the corpus to construct a label activation metric. Such a metric leverages historical test samples to adaptively align with the test distribution, enabling the selection of distribution-adaptive activated negative labels. By further exploring the activation information within the current testing batch, we introduce a more fine-grained, batch-adaptive variant. To fully utilize label activation knowledge, we propose an activation-aware score function that emphasizes negative labels with stronger activations, boosting performance and enhancing its robustness to the label number. Our TANL is training-free, test-efficient, and grounded in theoretical justification. Experiments on diverse backbones and wide task settings validate its effectiveness. Notably, on the large-scale ImageNet benchmark, TANL significantly reduces the FPR95 from 17.5% to 9.8%. Codes are available at [YBZh/OpenOOD-VLM](https://github.com/YBZh/OpenOOD-VLM).

1. Introduction

In open environments, artificial intelligence (AI) models inevitably encounter out-of-distribution (OOD) data, *i.e.*, samples outside predefined categories. Existing vision models often misclassify these OOD samples as known categories [53], posing significant safety issues. Therefore, accurately

detecting OOD samples is critical for AI safety.

Traditional OOD detection methods in the image domain primarily rely on visual features [20, 36, 44]. Recently, with the rapid development of vision-language models, leveraging textual knowledge to enhance OOD detection has gained increasing attention [14, 48, 69]. Among these, NegLabel [29] mines thousands of negative labels by identifying words with large cosine distances from in-distribution (ID) labels and detects OOD images by selecting those with higher cosine similarity to negative labels. Although NegLabel achieves impressive results, many negative labels present poor activation on a certain OOD test set, as shown in Fig. 1a. This limits their effectiveness and introduces noise, as these labels are closer to ID data and fail to capture the characteristics of OOD images effectively. Removing these less activated labels benefits the detection of OOD samples, as illustrated in Fig. 1b, highlighting their negative impact. This motivates us to explore negative labels with stronger activation on OOD samples to enhance OOD detection.

To this end, we propose Test-time Activated Negative Labels (TANL) for OOD detection. At the core of TANL is an activation metric that quantifies how “active” a particular class is across a dataset, measured by the average classification probability of a label across the associated images, as defined in Eq. 5. Using this metric, we dynamically evaluate the activation of each class in the corpus set across ID and OOD datasets, approximated using high-confidence positive and negative test images. This activation information captures the overall characteristics of the test distribution, enabling the selection of distribution-adaptive activated labels, which present high activation on negative images and low activation on positive images. Additionally, we explore the activation information within the current testing batch, resulting in a more fine-grained, batch-adaptive variant. To ensure stability at the beginning of testing, we initialize the activation scores using ID labels and noise images as positive and negative samples, respectively. Furthermore, considering the varying importance of different negative labels, as validated by their activation levels, we introduce an

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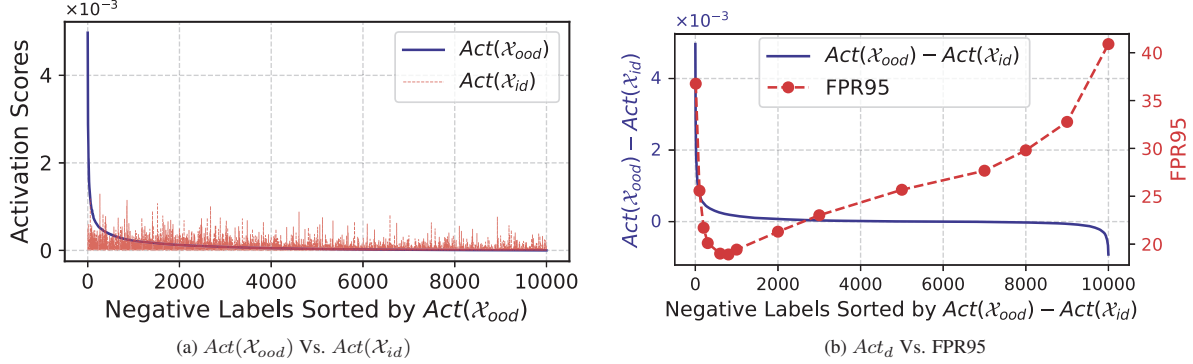


Figure 1. Activation analyses with negative labels mined in [29]. (a) Negative labels on a specific OOD dataset exhibit a long-tailed activation score distribution. Some labels activate more strongly on the ID dataset than on OOD, potentially misleading OOD detection. (b) A small subset of negative labels strongly activates on OOD, enabling effective detection. Most labels respond similarly across ID and OOD, slightly harming detection, while some activate higher on ID, significantly degrading performance. The FPR95 results are obtained with negative labels of top activations via Eq. 4. These analyses use ground truth labels from ImageNet (ID) and Places (OOD) datasets.

activation-aware scoring function for OOD detection that emphasizes labels with higher activation. This score function not only boosts the OOD detection performance but also enhances its robustness to the label number.

Extensive experiments are conducted to validate the effectiveness of our proposed methods. On the large-scale ImageNet dataset, our method reduces the FPR95 of the activation-agnostic NegLabel [29] by 15.6% and outperforms the current leading approach [6] by 7.7%. Unlike existing methods [29, 82], which typically introduce thousands of negative labels to cover activated labels but inevitably include less activated ones, our method specifically targets activated labels, achieving superior performance with a significantly reduced number of labels. Moreover, our approach is zero-shot, training-free, and test-efficient, demonstrating high scalability to different model backbones and robustness to near-OOD, full-spectrum OOD, and medical OOD settings. Theoretical analysis further explains its effectiveness. We summarize our contributions as follows:

- We introduce an activation metric to quantify how “active” a particular class is across a dataset and reveal that current methods employ less-activated negative labels for OOD datasets, which hinders the distinction of OOD samples. This motivates us to explore activated negative labels to enhance OOD detection.
- To this end, we explore activated negative labels by dynamically estimating the activation scores across the entire corpus dataset in the testing process. We introduce distribution-adaptive and batch-adaptive variants and a novel activation-aware scoring function to fully utilize the mined activation knowledge. Theoretical analysis is conducted to explain its effectiveness.
- We conduct extensive experiments to validate the proposed components. Our TANL outperforms the leading method by 7.7% in FPR95 on the large-scale ImageNet benchmark.

Our approach is zero-shot, training-free, test-efficient, and presents high scalability to different backbones and diverse task settings.

2. Related Work

OOD Detection aims to identify test samples with undefined semantic concepts, thereby enhancing the reliability of models in open environments. Classical OOD detection methods typically explore knowledge exclusively from the visual domain and can be roughly categorized into score-based [20, 26, 36, 43, 44, 61, 68, 70, 71], distance-based [11, 49, 60, 62–64], and generative-based [31, 58] approaches. Among these, score-based methods are the most popular, including score functions based on prediction confidence [20, 43, 61, 68, 71], additional discriminators [31], and energy score [44, 70].

Recently, with the rise of vision-language models [28, 56, 84], enhancing visual OOD detection by leveraging text knowledge has gained increasing attention [34, 38, 50, 72, 86]. ZOC [14] pioneered this direction by generating potential OOD labels with a learned captioner. Adapting the text branch with prompt tuning is a popular approach, where methods such as LoCoOp [51], LAPT [83], CLIPN [69], and LSN [54] introduce negative training samples via background features, image generation, additional training sets, and local croppings, respectively. Beyond single-text modality fine-tuning, multi-modal fine-tuning has been explored in [30] by jointly tuning both the text and visual branches.

Unlike these tuning-based methods, which typically require labeled training data, some approaches focus on training-free zero-shot pipelines [1, 4, 55]. For instance, MCM [48] detects OOD samples by analyzing the similarity distribution between test images and ID labels. EOE [4], NegLabel [29], and CSP [6] further advance this by introducing negative labels and utilizing the similarity between

test images and ID/negative labels to enhance OOD detection. While these methods achieve good results, we reveal that many of these negative labels present poor activation on OOD images, hindering the accurate detection of OOD samples, as shown in Fig. 1. To address this, we explore more activated negative labels and design a corresponding activation-aware score function, significantly reducing the FPR95 on the ImageNet dataset from 25.4%, achieved by the NegLabel, to 9.8%.

Test Time Adaptation enables models to dynamically update during testing to adapt to changes in test data [42], which has been recently introduced into OOD detection. Some methods [15, 17, 77] require test-time optimization, which, although achieving certain performance improvements, significantly reduces testing speed. Recently, some methods [78, 82] have also sought to avoid test-time optimization, achieving rapid adaptation to the environment. For example, OODD [78] caches high-confidence positive/negative samples in a dictionary and detects OOD samples based on the cosine similarity between the test samples and the cached features. AdaNeg [82] introduces adaptive negative proxies by combining negative labels and corresponding negative images. In contrast, we explore a new perspective of label activation and propose a novel activation-guided label mining strategy along with an activation-aware score, significantly boosting OOD detection.

3. Methods

3.1. Preliminaries

OOD Detection. Consider $\mathcal{X}$ as the image domain and $\mathcal{Y}^+ = \{y_1, \dots, y_C\}$ as the set of ID class labels, where $\mathcal{Y}^+$ consists of textual elements such as $\mathcal{Y}^+ = \{\text{cat}, \text{dog}, \dots, \text{bird}\}$, and C is the total number of classes. Let x^{in} and x^{ood} represent random variables corresponding to ID and OOD samples from $\mathcal{X}$, respectively. The marginal distributions for ID and OOD samples are denoted by $\mathcal{P}_x^{in}$ and $\mathcal{P}_x^{ood}$. In standard classification tasks, it is assumed that a test image x belongs to the ID distribution and is associated with a specific ID label, *i.e.*, $x \in \mathcal{P}_x^{in}$ and $y \in \mathcal{Y}^+$, where y is the label of x . However, in open environments, AI systems often encounter data from unknown classes, characterized by $x \in \mathcal{P}_x^{ood}$ and $y \notin \mathcal{Y}^+$. These OOD samples are typically misclassified as the known ID categories [53, 59], potentially resulting in unsafe decisions. To tackle such issues, OOD detection aims to reliably classify ID samples into their respective categories while rejecting OOD samples as non-ID. Classification among ID categories is done using a C -way classifier, following conventional methods [18, 33]. Meanwhile, OOD detection employs a scoring function S [36, 43, 44] to distinguish between ID and OOD inputs:

$$G_\gamma(x) = \begin{cases} \text{ID}, & \text{if } S(x) \geq \gamma; \\ \text{OOD}, & \text{otherwise,} \end{cases} \quad (1)$$

where G_γ is the OOD detector with a threshold $\gamma \in \mathcal{R}$ and $S(x)$ is a scoring function assigning higher scores to samples likely belonging to ID classes.

CLIP and NegLabel. For an ID test image x belonging to the label space $\mathcal{Y}^+$, we extract its image feature vector $v = f_{img}(x) \in \mathcal{R}^D$ and the text feature $t_i = f_{txt}(\rho(y_i)) \in \mathcal{R}^D$ using pre-trained CLIP encoders, where D denotes the feature dimension. The functions $f_{img}(\cdot)$ and $f_{txt}(\cdot)$ represent images and text encoders, respectively. The function $\rho(\cdot)$ serves as a text prompt mechanism, typically defined as ‘a photo of a <label>’, where ‘<label>’ corresponds to the actual class name such as ‘cat’ or ‘dog’. Both v and t_i are normalized using L_2 normalization along the dimension D . The zero-shot classification probabilities are then computed with the similarity between v and t_i :

$$p_i^{id} = \frac{\exp(vt_i)}{\sum_{j=1}^C \exp(vt_j)}, \quad (2)$$

where the temperature scaling factor is omitted for simplicity.

The CLIP model has been recently extended to OOD detection [29, 48, 83]. Specifically, Jiang *et al.* [29] introduce negative class labels $\mathcal{Y}^- = \{\tilde{y}_1, \dots, \tilde{y}_M\}$ by mining text labels distant from ID classes $\mathcal{Y}^+$ within an extensive corpus dataset $\mathcal{Y}^{cor} = \{\hat{y}_1, \dots, \hat{y}_N\}$:

$$\mathcal{Y}^- = \text{Top}(\{d_i\}_{i=1}^N, \mathcal{Y}^{cor}, M) \quad (3)$$

where d_i measures the cosine distance between $\hat{y}_i$ and ID label set $\mathcal{Y}^+$. M and N respectively denote the number of selected negative classes and all classes in the corpus dataset, and $N \geq M$. The operation $\text{TOP}(A, B, M)$ retrieves the indices of the top- M largest elements in set A and uses them to select the corresponding elements from set B . Sets $\mathcal{Y}^-$ and $\mathcal{Y}^+$ are disjoint, *i.e.*, $\mathcal{Y}^- \cap \mathcal{Y}^+ = \emptyset$. Then, the ID images can be detected as those with higher similarity to ID labels and lower similarity to negative ones, leading to the following score function:

$$S_{nl}(v) = \sum_{i=1}^C \frac{\exp(vt_i)}{\sum_{j=1}^C \exp(vt_j) + \sum_{j=1}^M \exp(v\tilde{t}_j)}, \quad (4)$$

where $\tilde{t}_i = f_{txt}(\rho(\tilde{y}_i)) \in \mathcal{R}^D$ is the text feature of mined negative label $\tilde{y}_i$.

3.2. Motivation: Label Activation Analyses

Although the negative labels described above perform well, they have a significant limitation that hinders their effectiveness. Specifically, the negative labels in Eq. 3 are derived solely from ID labels, without considering the test distribution in real-world applications. Consequently, many negative labels exhibit very low activation scores for a specific OOD test set and hinder the OOD detection, as shown in Fig. 1. To better understand this, we first introduce the concept of

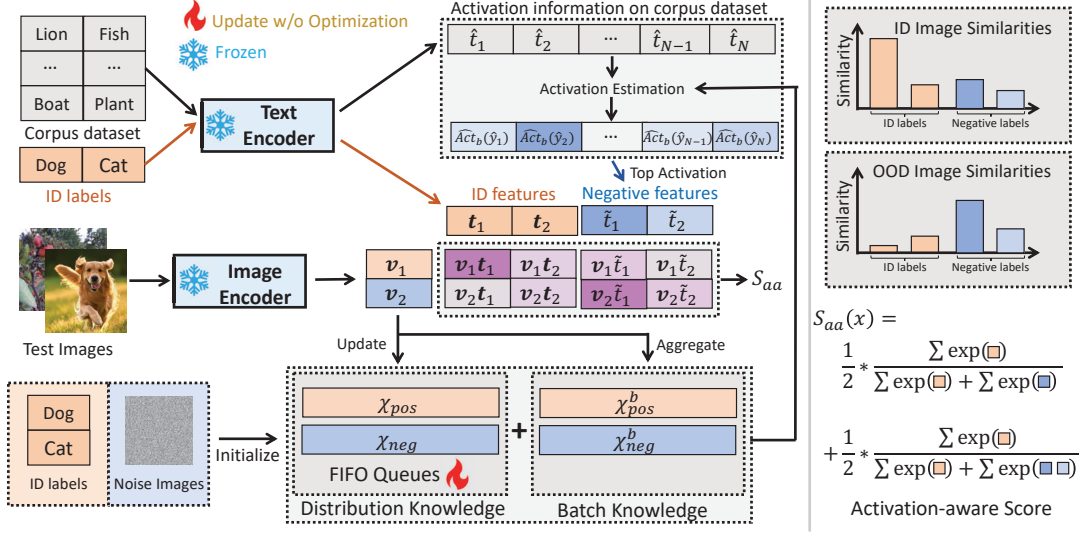


Figure 2. Overall framework of TANL. We dynamically explore activated negative labels from the corpus dataset in the testing process, where the activation information is measured based on the similarity between texts and the mined positive/negative images. The activation-aware score is illustrated as a simplified example of Eq. 15 with $M = 2$ and $C = 2$.

the activation score, which measures the average probability assignment of class $\hat{y}_i$ on a dataset $\mathcal{X}$:

$$Act(\mathcal{X}, \hat{y}_i) = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \frac{\exp(v\hat{t}_i)}{\sum_{j=1}^C \exp(v\hat{t}_j) + \sum_{j=1}^N \exp(v\hat{t}_j)}, \quad (5)$$

where $\hat{t}_i = f_{txt}(\rho(\hat{y}_i)) \in \mathcal{R}^D$. The $Act(\mathcal{X}, \hat{y}_i)$ reflects the average similarity between the class $\hat{y}_i$ and images in the dataset $\mathcal{X}$. A higher $Act(\mathcal{X}, \hat{y}_i)$ indicates greater similarity between the samples in $\mathcal{X}$ and class $\hat{y}_i$, and vice versa. An ideal negative label $\hat{y}_i$ should exhibit higher activation scores on OOD dataset (e.g., higher $Act(\mathcal{X}_{ood}, \hat{y}_i)$) and lower activation scores on ID dataset (e.g., lower $Act(\mathcal{X}_{id}, \hat{y}_i)$). However, as shown in Fig. 1a, although most negative labels derived in Eq. 3 exhibit low activation scores on the ID dataset, many simultaneously exhibit low activation scores on the OOD dataset—sometimes even lower than those observed on the ID dataset. These negative labels, which demonstrate lower activation scores on OOD samples, adversely impact OOD detection, as evidenced in Fig. 1b.

3.3. Our Approach

Distribution-adaptive Activated Labels. The above analyses on label activation motivate us to explore negative labels with higher activation on the OOD dataset and lower activation on the ID dataset, e.g., to find the negative labels that present high activation scores for OOD detection:

$$\mathcal{Y}^- = Top(\{Act_d(\hat{y}_i)\}_{i=1}^N, \mathcal{Y}^{cor}, M), \quad (6)$$

$$\text{where } Act_d(\hat{y}_i) = Act(\mathcal{X}_{ood}, \hat{y}_i) - Act(\mathcal{X}_{id}, \hat{y}_i). \quad (7)$$

However, in open environments, OOD data is generally unknown and may even change dynamically, making it difficult to obtain in advance. To address these problems, we propose a test-time adaptation strategy to dynamically evaluate the activation levels of candidate labels online, thereby selecting the most effective negative labels. Specifically, we approximate the activation score in Eq. 7 with cached positive and negative images:

$$\widehat{Act}_d(\hat{y}_i) = Act(\mathcal{X}_{neg}, \hat{y}_i) - Act(\mathcal{X}_{pos}, \hat{y}_i), \quad (8)$$

where $\mathcal{X}_{neg}$ and $\mathcal{X}_{pos}$ are fixed-length first-in-first-out (FIFO) queues that are dynamically updated with high-confidence positive and negative samples, respectively:

$$\begin{aligned} \mathcal{X}_{pos} &= \text{Update}(\mathcal{X}_{pos}, \{v \in \mathcal{B} \mid S_{aa}(v) \geq \gamma + (1 - \gamma)g\}, L), \\ \mathcal{X}_{neg} &= \text{Update}(\mathcal{X}_{neg}, \{v \in \mathcal{B} \mid S_{aa}(v) < \gamma - \gamma g\}, L), \end{aligned} \quad (9)$$

where $\gamma \in [0, 1]$ is the threshold distinguishing ID and OOD samples, $g \in [0, 1]$ is the gap used to filter high-confidence samples, $\mathcal{B}$ indicates the test image batch, and L is the capacity of the queue. S_{aa} is our proposed activation-aware score function, as defined in Eq. 15. We construct $\mathcal{X}_{pos/neg}$ with image features v , equivalent to vanilla images and more compact. This FIFO queue design ensures that the activation score remains responsive to environmental changes by focusing on the most recent high-confidence samples. It effectively filters outdated information, allowing the model to adapt dynamically and maintain robust performance in evolving scenarios.

To ensure a stable start for our method during testing, we respectively initialize $\mathcal{X}_{pos}$ and $\mathcal{X}_{neg}$ with features of ID labels and noise images:

$$\begin{aligned}\mathcal{X}_{pos} &= \text{Sampling}(\{\mathbf{t}_i\}_{i=1}^C, L) \\ \mathcal{X}_{neg} &= \{f_{img}(\mathbf{x}_i^{noise})\}_{i=1}^L,\end{aligned}\quad (10)$$

where $\mathbf{x}_i^{noise}$ is the image with random Gaussian noises and $\text{Sampling}(A, L)$ represents the operation of randomly selecting L elements from set A . We initialize the positive image set $\mathcal{X}_{pos}$ using ID label features, considering the alignment between text and image features in the shared feature space. This initialization helps construct the initial activated labels and the corresponding S_{aa} score function, providing a stable foundation for subsequent updates.

Batch-Adaptive Activated Labels. The above method enhances OOD detection by utilizing activation information from historical test samples. However, in addition to historical samples, the activation information in the current testing batch is also critical, as it captures the self-characteristics of test instances, especially under scenarios with temporal shifts. To address this, we propose a more fine-grained, batch-adaptive variant by incorporating the activation information within the current testing batch:

$$\begin{aligned}\mathcal{X}_{pos}^b &= \{\mathbf{v} \in \mathcal{B} \mid S_{aa}(\mathbf{v}) \geq \gamma + (1 - \gamma)g\}, \\ \mathcal{X}_{neg}^b &= \{\mathbf{v} \in \mathcal{B} \mid S_{aa}(\mathbf{v}) < \gamma - \gamma g\},\end{aligned}\quad (11)$$

where $\mathcal{X}_{pos}^b$ and $\mathcal{X}_{neg}^b$ represent the possible positive and negative samples in the testing batch $\mathcal{B}$. We then define the batch-adaptive activation score as:

$$\widehat{Act}_b(\hat{y}_i) = Act_b(\mathcal{X}_{neg}, \hat{y}_i) - Act_b(\mathcal{X}_{pos}, \hat{y}_i), \quad (12)$$

where

$$Act_b(\mathcal{X}_{pos}, \hat{y}_i) = \begin{cases} \alpha Act(\mathcal{X}_{pos}, \hat{y}_i) + (1 - \alpha) Act(\mathcal{X}_{pos}^b, \hat{y}_i) & \text{if } |\mathcal{X}_{pos}^b| > 0, \\ Act(\mathcal{X}_{pos}, \hat{y}_i) & \text{if } |\mathcal{X}_{pos}^b| = 0 \end{cases} \quad (13)$$

$$Act_b(\mathcal{X}_{neg}, \hat{y}_i) = \begin{cases} \alpha Act(\mathcal{X}_{neg}, \hat{y}_i) + (1 - \alpha) Act(\mathcal{X}_{neg}^b, \hat{y}_i) & \text{if } |\mathcal{X}_{neg}^b| > 0, \\ Act(\mathcal{X}_{neg}, \hat{y}_i) & \text{if } |\mathcal{X}_{neg}^b| = 0 \end{cases} \quad (14)$$

where $\alpha \in [0, 1]$ balances activation information from historical samples and the current batch, and $|\cdot|$ measures the set size.

Activation-aware Score. Considering that the activation scores of different negative labels vary, *i.e.*, their importance differs, we introduce for OOD detection the following score

Algorithm 1 Test-time Activated Negative Labels (TANL)

Require: ID labels $\mathcal{Y}^+$, an external corpus dataset $\mathcal{Y}^{cor}$ and a test set $\mathcal{X}_{test}$

- 1: Constructing the FIFO queues $\mathcal{X}_{pos}$ and $\mathcal{X}_{neg}$, and initializing them with Eq. 10.
 - 2: **for** Image batch $\mathcal{B} \in \mathcal{X}_{test}$ **do**
 - 3: Collecting positive set $\mathcal{X}_{pos}^b$ and negative set $\mathcal{X}_{neg}^b$ in batch $\mathcal{B}$ using Eq. 11
 - 4: Calculating the batch-adaptive activation score $\widehat{Act}_b(y_i)$ using Eq. 12
 - 5: Selecting the adaptive and activated negative labels $\mathcal{Y}^-$ using Eq. 6
 - 6: Getting the OOD detection score S_{aa} using Eq. 15
 - 7: Updating the FIFO queues using Eq. 9
 - 8: **end for**
 - 9: **Return** Collected final scores $\{S_{aa}\}$
-

to implicitly assign higher weights to those with higher activation scores:

$$S_{aa}(\mathbf{v}) = \frac{1}{M} \sum_{m=1}^M \sum_{i=1}^C \frac{\exp(\mathbf{v}\mathbf{t}_i)}{\sum_{j=1}^C \exp(\mathbf{v}\mathbf{t}_j) + \sum_{j=1}^m \exp(\mathbf{v}\tilde{\mathbf{t}}_j)}. \quad (15)$$

Recalling that the activated labels are selected based on ranked activation scores in Eq. 6, the ranking ensures that $\widehat{Act}_b(\tilde{y}_i) \geq \widehat{Act}_b(\tilde{y}_j)$ if $i < j$. This guarantees that labels with stronger activation (*e.g.*, $\tilde{\mathbf{t}}_j$ with smaller j) dominate the score, as their contributions are amplified by their repeated occurrence in the denominator. We find that this method not only improves OOD detection performance but also significantly enhances its robustness to the number of negative labels, as verified in Fig. 3a. We also analyze other activation-score variants in Appendix A6.5, where the negative labels are explicitly weighted according to their activation scores. While these variants also effectively utilize activation information, S_{aa} offers advantages in robustness and is adopted as the default. The overall framework is shown in Fig. 2 and summarized in Algorithm 1.

Theoretical Insight. We follow [29] to conduct the theoretical analysis from the perspective of multi-label classification and model the partial derivative of FPR_λ with respect to the number M of negative labels as:

$$\frac{\partial FPR_\lambda}{\partial M} = \frac{1}{2} \cdot \frac{\partial \text{erf}(z)}{\partial z} \cdot \frac{\partial z}{\partial M} = \frac{e^{-z^2}}{2\sqrt{2\pi}} \cdot \frac{p_1 - p_2}{\sqrt{Mp_2(1 - p_2)}}, \quad (16)$$

where $p_1 = P(\text{sim}(\mathbf{x}_{id}, \tilde{y}_i) \geq \psi | f, \mathbf{x}_{id}, \tilde{y}_i)$ indicates the average probability of classifying the ID image $\mathbf{x}_{id}$ as the negative label $\tilde{y}_i$, $\text{sim}(\mathbf{x}_{id}, \tilde{y}_i)$ represents the similarity between $\mathbf{x}_{id}$ and $\tilde{y}_i$, and ψ is a threshold. p_2 is similarly defined with the OOD image $\mathbf{x}_{ood}$. $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$

Table 1. OOD detection results with ImageNet-1k, where a ViT-B/16 CLIP encoder is adopted.

Methods	OOD datasets									
	INaturalist		Sun		Places		Textures		Average	
	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓
Methods requiring training (or fine-tuning)										
ZOC [14]	86.09	87.30	81.20	81.51	83.39	73.06	76.46	98.90	81.79	85.19
CLIPN [69]	95.27	23.94	93.93	26.17	92.28	33.45	90.93	40.83	93.10	31.10
LoCoOp [51]	96.86	16.05	95.07	23.44	91.98	32.87	90.19	42.28	93.52	28.66
LAPT [83]	99.63	1.16	96.01	19.12	92.01	33.01	91.06	40.32	94.68	23.40
NegPrompt [40]	98.73	6.32	95.55	22.89	93.34	27.60	91.60	35.21	94.81	23.01
KR-NFT [86]	99.62	0.82	96.15	17.83	92.64	36.12	91.96	36.38	95.09	22.79
CMA [30]	99.62	1.65	96.36	16.84	93.11	27.65	91.64	33.58	95.13	19.93
AdaND [5]	99.49	1.91	95.49	20.53	94.55	20.95	93.01	21.76	95.58	16.00
SynOOD [39]	99.57	1.57	95.82	20.46	93.37	12.12	95.29	22.94	97.01	14.27
Training-free & non-adaptive methods										
MCM [48]	94.59	32.20	92.25	38.80	90.31	46.20	86.12	58.50	90.82	43.93
CoVer [79]	95.98	22.55	93.42	32.85	90.27	40.71	90.14	43.39	92.45	34.88
CMA [37]	96.89	23.84	93.69	30.11	93.17	29.86	88.47	47.35	93.05	32.79
EOE [4]	97.52	12.29	95.73	20.04	92.95	30.16	85.64	57.53	92.96	30.09
NegLabel [29]	99.49	1.91	95.49	20.53	91.64	35.59	90.22	43.56	94.21	25.40
DualCnst [34]	99.70	0.98	95.66	18.13	91.81	31.77	91.73	34.91	94.72	21.45
CSP [6]	99.60	1.54	96.66	13.66	92.90	29.32	93.86	25.52	95.76	17.51
Training-free & test-time adaptation methods										
AdaNeg [82]	99.71	0.59	97.44	9.50	94.55	34.34	94.93	31.27	96.66	18.92
Oodd [78]	99.79	0.85	97.17	12.94	92.51	30.68	94.51	30.67	96.00	18.79
TANL (Ours)	99.84	0.42	99.07	3.53	95.87	21.90	97.11	13.38	97.97	9.81

Table 2. OOD detection results under OpenOOD setting, where ImageNet-1k is adopted as ID dataset. Full results are available in Tab. A7.

Methods	FPR95 ↓		AUROC ↑		ACC ↑ ID
	Near-OOD	Far-OOD	Near-OOD	Far-OOD	
Methods requiring training (or fine-tuning)					
AugMix [23] + ReAct [61]	—	—	79.94	93.70	77.63
SCALE [74]	—	—	81.36	96.53	76.18
AugMix [23] + ASH [10]	—	—	82.16	96.05	77.63
LAPT [83]	58.94	24.86	82.63	94.26	67.86
CMA [30]	56.25	15.29	84.46	96.47	82.64
Training-free & non-adaptive methods					
MCM [48]	79.02	68.54	60.11	84.77	66.28
NegLabel [29]	69.45	23.73	75.18	94.85	66.82
Training-free & test-time adaptation methods					
AdaNeg [82]	67.51	17.31	76.70	96.43	67.13
TANL (Ours)	60.06	17.21	84.53	96.43	66.82

is the error function and $z = \sqrt{\frac{p_1(1-p_1)}{p_2(1-p_2)}} \text{erf}^{-1}(2\lambda - 1) + \frac{\sqrt{M}(p_1 - p_2)}{\sqrt{2p_2(1-p_2)}}$. As shown in Eq. 16, to reduce the FPR_λ by increasing the number M of negative labels, a prerequisite is that $p_1 - p_2 < 0$, suggesting that $\text{sim}(\mathbf{x}_{id}, \tilde{y}_i) < \text{sim}(\mathbf{x}_{ood}, \tilde{y}_i)$ holds on average. In other words, the negative

label $\tilde{y}_i$ should have higher similarity with the OOD samples and lower similarity with the ID samples.

There is a close relationship between the theoretical objective in Eq. 16 and our algorithm implementation in Eq. 7. Specifically, $\text{Act}(\mathcal{X}_{ood}, \hat{y}_i)$ measures the activation level of candidate label $\hat{y}_i$ on the OOD dataset by averaging the nor-

Table 3. Full-spectrum OOD detection results under the OpenOOD setting, where ImageNet-1k, ImageNet-C, ImageNet-R, ImageNet-V2 are used as ID datasets. Full results are shown in Tab. A8.

Methods	FPR95 ↓		AUROC ↑	
	Near-OOD	Far-OOD	Near-OOD	Far-OOD
Methods requiring training (or fine-tuning)				
AugMix [23] + SHE [80]	84.45	60.26	69.66	83.06
LSA [46]	70.56	48.06	78.22	86.85
ISH + SCALE [74]	—	—	68.04	89.46
LAPT [83]	71.18	33.07	74.77	92.14
Training-free & non-adaptive methods				
MCM [48]	85.37	69.87	58.97	77.11
NegLabel [29]	76.25	33.30	72.77	92.02
Training-free & test-time adaptation methods				
TANL (Ours)	68.71	22.48	78.90	94.35

malized similarity between $\hat{y}_i$ and OOD images; similarly, $Act(\mathcal{X}_{id}, \hat{y}_i)$ reflects the normalized similarity between $\hat{y}_i$ and ID images. According to Eq. 6, we select the ideal negative labels $\{\hat{y}_i\}_{i=1}^M$ that exhibit higher activation with OOD samples while maintaining lower activation with ID samples, explicitly ensuring $p_1 - p_2 < 0$ in Eq. 16.

4. Experiments

4.1. Setup

Datasets. We primarily conducted experiments using ImageNet-1k [8] as the ID dataset. We adopted four OOD datasets [7, 65, 73, 85] following common practice [26, 29, 48] and also performed experiments under the OpenOOD setting [75, 81]. Additionally, we validated our method under the full-spectrum setting [76], with a smaller ID dataset of CIFAR [32], and with medical images [66, 81]. More details are provided in Appendix A6.1.

Implementation Details. We employ the visual encoder of ViT/16, pretrained with CLIP [56], and investigate other backbone architectures in Tab. A12. Following the design of NegLabel, we use the text prompt “The nice <label>” and set the temperature parameter to 0.01. We set the number of negative labels to $M = 1000$ in Eq. 15, the gap $g = 0.2$, and $L = 300$ in Eq. 9, and $\alpha = 0.95$ in Eq. 12 based on analysis on ImageNet. These hyperparameters were fixed and work well on other datasets. As shown in Fig. A5b, an automatically determined threshold γ generally performs comparably to manually searched ones. We adopt the evaluation metrics of FPR95, AUROC, and ID ACC, following standard protocols [26, 29, 48].

4.2. Main Results

ImageNet Results. As shown in Tab. 1, our TANL significantly outperforms existing training-free methods and even surpasses techniques that require additional training. Detailed discussions are provided in Sec. A6.2 and complete comparisons are presented in Tab. A6.

ImageNet Results with OpenOOD Setup. As shown in Tab. 2, TANL surpasses existing zero-shot training-free

Table 4. Ablation analyses, where results (FPR95↓) are reported with ImageNet ID dataset under the OpenOOD setup. “Dis-adapt”, “Batch-adapt”, and “AAScore” represent the distribution-adaptive activated score in Eq. 8, batch-adaptive variant in Eq. 11, and activation-aware score in Eq. 15, respectively.

Dis-adapt	Batch-adapt	AAScore	Near-OOD	Far-OOD
NegLabel Baseline			69.45	23.73
✓			61.57	17.52
	✓		60.85	17.25
✓		✓	60.59	17.27
	✓	✓	60.06	17.21

methods and achieves performance comparable to training-required competitors. Notably, it outperforms the close competitors [29, 82], especially in the challenging near-OOD setting, demonstrating the effectiveness of activated negative labels. Our method also preserves ID classification accuracy by freezing the pre-trained CLIP model.

Full-spectrum OOD Detection. As shown in Tab. 3, TANL not only achieves high distinguishability against semantic shifts but also exhibits strong robustness to covariate shifts. **More results** on CIFAR and medical datasets are provided in Tab. A9 and Tab. A11, respectively.

4.3. Analyses and Discussions

Ablation. As shown in Tab. 4, adopting activated negative labels significantly outperforms NegLabel, which uses fixed and activation-agnostic labels, verifying the importance of label activation. Additionally, incorporating current batch information brings a slight improvement and adopting activation-aware score consistently leads to an advantage.

Label Number and S_{aa} . As illustrated in Fig. 3a, utilizing activated negative labels consistently outperforms NegLabel, validating the importance of label activation. Specifically, when the number of negative labels M is small, the selected labels present higher activation scores, significantly reducing the OOD detection error. As M increases, less-activated labels are gradually included, resulting in a gradual performance decline, consistent with the theoretical analysis in Sec. 3.3. Using the same score function S_{nl} , our TANL (w/o S_{aa}) consistently outperforms NegLabel across different numbers of negative labels, fairly validating the advantage of our label selection strategy. Additionally, TANL consistently outperforms TANL (w/o S_{aa}), particularly when the number of negative labels is large. This confirms that incorporating label activation information into the score function enhances its robustness to the number of negative labels.

Negative Label Selection Criterion. As shown in Fig. 3b, mining negative labels with the activation information of negative samples significantly outperforms using positive information alone. Combining both positive and negative information yields the best results.

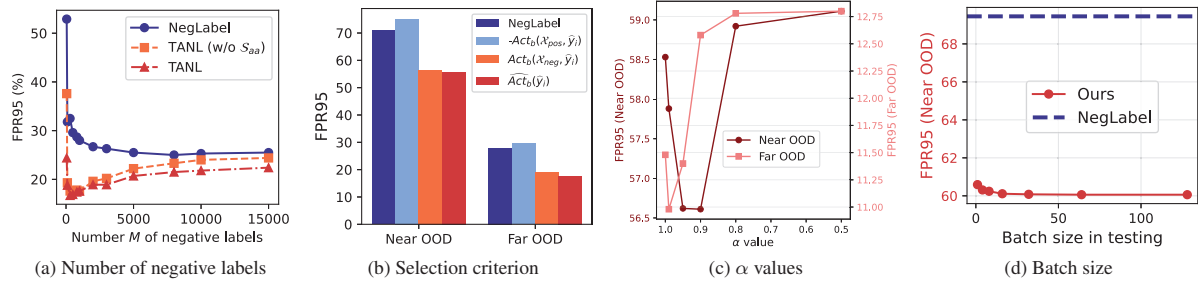


Figure 3. Analyses on (a) number M of selected negative labels, (b) selection criterion of negative labels, (c) α values, and (d) batch size under OpenOOD setting.

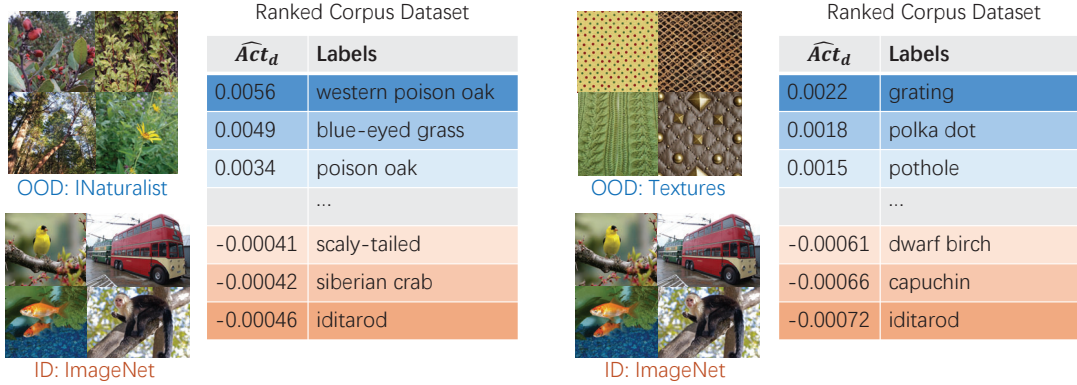


Figure 4. Visualization of ranked corpus dataset, where candidate labels with higher activation scores are utilized.

α Analyses. As shown in Fig. 3c, reducing the α values from 1.0 to 0.95 brings certain improvements, validating the effectiveness of batch knowledge. However, further decreasing the α value leads to overfitting to batch information, which in turn harms OOD identification. $\alpha = 0.95$ achieves a good balance between distribution and batch information and is adopted as the default setting.

Batch Size. As shown in Fig. 3d, larger batch sizes improve performance, but our method remains effective even with a batch size of 1, confirming its practicality in streaming applications.

Visualization. We visualize the activated labels in Fig. 4. The selected activated labels exhibit a high degree of similarity with the OOD dataset. More discussion and visualization are available in Sec. A6.5.

Complexity Analyses. As shown in Table 5, our approach achieves superior performance with moderate test speed. We also observe that our method can achieve strong results even with smaller backbones (e.g., ResNet50), offering advantages in both speed and accuracy.

More discussions on the queue length L , threshold γ , and gap g in Eq. 9, different VLM backbones, various corpus datasets, activation metric, robustness to temporal shift and early errors, and limitations are provided in Appendix A6.5.

Table 5. Time complexity analyses. ‘Training’ measures the training time, and ‘Param.’ presents the number of learnable parameters. ‘FPS’ reflects the inference speed with a batch size of 256.

Methods	Backbones	Training	FPS $\uparrow$	Param.	FPR95 $\downarrow$
ZOC [14]		>24h	287	336M	85.19
CLIPN [69]		>24h	605	37.8M	31.10
LoCoOp [51]	ViT/16	9h	625	8K	28.66
MCM [48]		–	625	–	43.93
NegLabel [29]		–	592	–	25.40
AdaNeg [82]		–	476	–	18.92
TANL (Ours)		–	527	–	9.81
TANL (Ours)	ResNet50	–	790	–	10.37

5. Conclusion

In this paper, we proposed a novel OOD detection method that dynamically explored activated negative labels in a test-time manner. We also designed an activation-aware score function to fully utilize the mined activation knowledge. Our approach was training-free, test-efficient, and grounded in theoretical justification. Additionally, it demonstrated high scalability to different model backbones and robustness to near-OOD, full-spectrum OOD, and medical OOD settings. We hope this work draws attention to the activation information in the OOD detection community.

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